

RACHIT CHADHA

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Education

Georgia Institute of Technology

Master of Science in Computer Science, Machine Learning Track

Atlanta, Georgia

University of Maryland

Bachelor of Science in Information Science, Data Science Concentration

College Park, Maryland

Relevant Coursework

- Data Science
- Machine Learning
- Computer Vision
- Big Data Systems
- Algorithms
- AI
- Deep Learning
- NLP
- Statistics
- Data Visualization

Technical Skills

Languages: Python, Pyspark, Scala, SQL, R, JavaScript
Frameworks: PyTorch, Scikit-Learn, Pandas, NumPy, Statsmodels, LangChain, Vector Databases
Machine Learning: Time-Series Forecasting (ARIMA, Prophet, XGBoost), Regression, Classification, Causal Inference
Data Engineering: Databricks, MongoDB, Azure, Data Lakes, ETL, Docker, Snowflake, Power BI, Tableau, Git, AWS
Certificates: Asset Management Professional (IAM Certified), Hackathon Winner (2024), Hackathon Judge (2025)

Experience

Senior Data Scientist

May 2024 – Present

Turner & Townsend

New York City, NY

- Leading data team and leveraging cross-functional teams to convert analytical insights into production-grade tools.
- Designed and deployed end-to-end forecasting and anomaly detection pipelines in Azure Databricks using PySpark and SQL, integrating sensor and operational data to detect system anomalies and forecast risks with 25% higher accuracy.
- Developed interpretable predictive models (XGBoost, LightGBM) improving outage forecast and risk accuracy by 20%.
- Streamlined Oracle SQL pipelines with Maximo data lineage and quality monitoring, cutting execution time by 15%.
- Built and automated Power BI dashboards for real-time KPIs used by operations teams across multiple sites.
- Engineered batch process pipeline to pull images and analyze contents from Maximo via custom Python-SQL ETL.

Graduate Assistant, Research and Lecture

Aug 2024 – Dec 2024

Georgia Institute of Technology

Atlanta, GA

- Served as Graduate Teaching Assistant for CSE 6242 Data & Visual Analytics, supporting more than 200 students in applied data science, visualization, SQL, machine learning, and cloud-based analytics assignments.
- Led weekly office hours and technical support sessions, helping students troubleshoot Python, SQL, D3/JavaScript, machine learning, AWS, and data visualization implementation issues.
- Designed and tested Azure, SQL, and DataBricks based assignments on data visualization and modeling workflows.

Consultant, Data Engineering & Analytics

Apr 2022 – Jun 2023

AMCL

New York City, NY

- Developed time-series forecasting models (ARIMA, XGBoost) to predict equipment degradation and reliability trends.
- Automated data ingestion and transformation pipelines in Python + SQL, enabling real-time anomaly detection.
- Modeled real-time environmental and sensor data to forecast degradation patterns for optimizing equipment reliability.
- Applied A/B testing and causal inference to quantify policy impacts, supporting data-driven infrastructure planning.

Project Engineer, Analytics

Aug 2020 – Mar 2022

Aegis Project Controls

Washington, DC

- Implemented time-impact risk models using SQL, Python, and simulation-based forecasting for operational efficiency.
- Built and maintained ETL pipelines and Power BI dashboards for procurement and supply chain analytics.

Intel

May 2019 – Aug. 2019

Analytics Intern, eSports

Singapore

- Presented analytical insights to support a data-centric strategy for Intel Extreme Masters Sydney 2020, targeting expanded engagement with the Asia-Pacific eSports audience.
- Used Python, Twitch API, and Power BI to scrape, analyze, visualize, and present APAC eSports market trends and engagement data.
- Collaborated with UX, marketing, APJ/ANZ team to audit Intel digital marketing content across 13 specialty accounts.
- Supported A/B testing and content optimization using CSS, JavaScript, SEO analysis, and digital performance review.
- Prepared stakeholder-facing summaries connecting gaming audience behavior, digital engagement metrics, and event marketing opportunities.

Projects & Research

Hackathon Winner: Real-time Geospatial Energy Forecast App | *Python, MongoDB, Spark, Mapbox.js, Node.js, AWS*

- Real-time tool for forecasting tidal power using spatial modeling, hackathon winner amongst 1100+ participants.
- Ingested and processed NOAA buoy data; performed modeling for energy cost mitigation and emissions reduction. [Link](#)

Efficient LoRA with Sparse Attention Finetuning | *PyTorch, LoRA, LLaMA-3, Mistral, GPU Optimization*

- Co-authored a novel PEFT method for LLMs reducing adapter size by **75%**, achieving a **23% speedup** in inference.
- Built calibration-based finetuning framework reducing GPU memory usage by **20%**; optimized attention sparsity.
- Evaluated across multiple tasks to benchmark generalizability and attention efficiency. [Link](#)

LLM for C Programming Challenges | *PyTorch, LLaMA-2, LoRA, QLoRA, CFG, Encoders, NLP*

- Fine-tuned transformer models on curated LeetCode C programming tasks with quantization and control flow graphs.
- Achieved **83%** solution success rate and reduced compilation errors by **21%** with optimized GPU performance. [Link](#)

Product Growth Study: Payouts and Forecasting | *Time-series Prophet model, Pandas, NumPy, Seaborn*

- Built and validated a time series forecasting model to project payment volumes and detect anomalies in seasonality.
- Modeled payout variance and confidence intervals to inform operations and payment cadence optimization. [Link](#)

Boiler Asset Intelligence & Predictive Overhaul Forecasting Platform | *Oracle SQL, IBM Maximo, Python, Power BI*

- Developed a predictive maintenance analytics solution for boiler overhauls, integrating PM, IN, and CM work orders with parent-child Maximo logic, priority classification, weather impact analysis, and heating degree day context.
- Created forecasting models, SQL logic, DAX measures, and Power BI dashboard pages to monitor overhaul completion, remaining work, priority trends, and future maintenance outlooks.

User Engagement Analytics for Video Streaming Platform | *Python, NumPy, Pandas, Scikit-learn, Machine Learning*

- Applied EDA and recommendation systems on user behavior metrics (playback, churn & bounce rate) using statistical models such as trend analysis and descriptive statistics leveraging Matrix Factorization, SVD, K-means, LightGBM.

DataCleaner | *Pandas, NumPy, scikit-learn, SciPy, Seaborn*

- Developed an easy-to-use preprocessing tool for automated data cleaning and interestingness-based pruning in machine learning workflows.
- Published a research paper describing the tool's approach to handling missing data, outliers, encoding, and feature preparation.